Systems Programming Project:
Corrected Technical Validation Report

Integrated System Testing

February 1, 2026

1 Executive Summary

This report provides a critical assessment of the integrated system comprising Labs 1 through 5. While the core virtual machine and garbage collection logic are functional, automated testing has identified significant deficiencies in debugger synchronization, parser-to-VM mapping, and the implementation of advanced shell features.

2 Identified Failures and Logic Gaps

2.1 Debugger and Parser Misalignment

Tests 14 and 15 (Breakpoint and Continue) consistently failed with the error `"No instruction found at line X"`.

- **Cause:** There is a fundamental misalignment between the Parser's line-counting metadata and the VM's bytecode mapping. If the Parser miscounts a newline, the Debugger cannot resolve source-code locations to bytecode offsets.
- **Impact:** Mandatory requirements for setting and removing software breakpoints are currently unreliable.

2.2 Output Synchronization Failures

Phase 5 of the robustness suite and several automated tests (Tests 6, 29, 50) failed due to missing expected strings like `"Debugger started"` or `"Exiting system"`.

- **Cause:** The C program's output buffer is not being flushed immediately (`fflush(stdout)`).
- **Impact:** The automated validation logic fails because it cannot read the system state before the process terminates.

2.3 Memory Reporting Discrepancies

Tests 31 and 44 failed because the system output `"Heap:  X objects"` instead of the strictly required `"Active Objects:  X"`.

- **Impact:** Automated regression tests failed despite underlying memory logic being potentially correct.

3 Untested and Unmet Requirements

3.1 Shell Features (Lab 1)

The following features, while defined in the documentation, have not been fully verified:

- **Multi-Stage Pipelines:** Multi-stage pipelines (e.g., `ls | grep | sort`) have not been stress-tested in the integrated environment.
- **Redirection:** Input (`<`) and output (`>`) redirection remain untested for edge cases like missing files.
- **Background Execution:** The `&` symbol for background tasks was not exercised in the 50-test suite.

3.2 Parser and VM (Labs 3–4)

- **Math Expression Ambiguity:** The parser initially failed to handle complex parenthetical math (e.g., `(5+5)*2`) because the grammar lacked specific rules for parentheses.
- **Error Recovery:** The system currently fails to meet the optional "Error Recovery" extension; it stops at the first error found rather than continuing to parse the rest of the file.

4 Compliance Matrix

Lab	Requirement	Status	Notes
Lab 1	SIGINT Management	MET	Shell ignores Ctrl-C.
Lab 2	CPU Register Inspection	PARTIAL	Shows VM registers, not x86 registers.
Lab 4	Function Call Frames	MET	`CALL` and `RET` are functional.
Lab 5	Cyclic Reference GC	MET	Tracing handles cycles correctly.

5 Conclusion

The system meets the core functional requirements for memory management and bytecode execution. However, the failures in debugger interaction and shell command parsing indicate that the integration is not yet robust enough for production-level stability.